

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition? Answer Choices: A. It activates TLR5 on the surface of macrophages. B. It primarily induces the Th1-cell response. C. Peptidoglycan is its major antigen that induces an immune response. D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think> The patient's symptoms, physical examination findings, and X-ray results are consistent with community-acquired pneumonia. The patient's response to cephalexin, an antibiotic, suggests that the causative agent is likely a bacterial species, such as Streptococcus pneumoniae, which is common in community-acquired pneumonia cases. Streptococcus pneumoniae is a gram-positive bacterium and its cell wall contains peptidoglycan. Peptidoglycan is a major antigen that activates the innate immune response by activating Toll-like receptors (TLRs), particularly TLR2. The TLR2 ligand, peptidoglycan, can induce both Th1 and Th17 cell responses but the initial activation of macrophages and neutrophils is through TLR2. The other options are less specific for the type of immune response elicited by Streptococcus pneumoniae. Encapsulation is a strategy used by some bacteria to evade the immune response, but it is not the primary immune response elicited by Streptococcus pneumoniae. </think> <answer> C. Peptidoglycan is its major antigen that induces an immune response. </answer>

